Module 2 Lab 2: Textbook Problem 4.2

Multiplication the Long Way

Course: CS 82A - Introduction to Data Science Assignment: Module 2 Lab 2

In [36]:
```
def multiply_for(a, b):
    result = 0

    for i in range(b):
        result = result + a

    return result
```

In [37]:
```
print("7 x 6 =", multiply_for(7, 6))
print("5 x 4 =", multiply_for(5, 4))
print("8 x 3 =", multiply_for(8, 3))
```

```
7 x 6 = 42
5 x 4 = 20
8 x 3 = 24
```

In [38]:
```
def multiply_while(a, b):
    result = 0
    count = 0

    while count < b:
        result = result + a
        count = count + 1

    return result
```

In [39]:
```
print("7 x 6 =", multiply_while(7, 6))
print("5 x 4 =", multiply_while(5, 4))
print("8 x 3 =", multiply_while(8, 3))
```

```
7 x 6 = 42
5 x 4 = 20
8 x 3 = 24
```

How the While Loop Stops

The while loop stops when the count is no longer less than b. Each time the loop runs, count increases by 1. When count reaches b, the condition becomes False, and the loop ends.

In [40]:
```
for a, b in [(7, 6), (5, 4), (8, 3)]:
    print(
        "For:", multiply_for(a, b),
```

```
        "| While:", multiply_while(a, b),
        "| Match:", multiply_for(a, b) == multiply_while(a, b)
    )
```

```
For: 42 | While: 42 | Match: True
For: 20 | While: 20 | Match: True
For: 24 | While: 24 | Match: True
```

AI Use Appendix

I used ChatGPT to translate and explain the assignment instructions from English into Burmese so I could understand what the assignment required. ChatGPT helped me understand the tasks and the steps needed to complete the assignment. I used the explanations in Burmese to guide my work.